

Lecture 9: Risk Aversion vs Intertemporal Substitution

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Intro

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Building Blocks

What is Risk Aversion?

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- You don't need the concept of *time* to think about risk aversion

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- X : a set of outcomes; $\Delta(X)$: the set of lotteries over X (the simplex)
- Let $\succeq$ be a preference relation over $\Delta(X)$
- You are **risk-averse** if, for any $p \in \Delta(X)$, you have $\delta_{\mathbb{E}[p]} \succeq p$

What Von-Neumann and Morgenstern Taught Us

- Typical assumptions: $\succeq$ is complete, transitive, and continuous
- Independence axiom: if $p \succeq q$, then for any r and $\alpha \in (0, 1)$, we have $\alpha p + (1 - \alpha)r \succeq \alpha q + (1 - \alpha)r$

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- Under independence, $\succeq$ can be represented by a utility function $U : \Delta(X) \rightarrow \mathbb{R}$ such that

$$U(p) = \sum_{x \in X} \pi(x)u(x)$$

where $\pi(x)$ is the probability of outcome x and $u(x)$ is the utility of outcome x

- The preferences we will cover in this lecture will violate exactly this axiom, stay tuned!

More or Less Risk Averse?

- The **Certainty Equivalent** of $p \in \Delta(X)$ is the amount of money c such that $\delta_c \sim p$

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- John is (weakly) **more risk-averse** than Mary if $CE_J(p) \leq CE_M(p)$ for all $p \in \Delta(X)$
- Pratt showed that this is all about how concave u is!
- Concavity is measured by second derivatives... but these are not scale-free!
- We typically measure risk aversion by $-u''(\cdot)/u'(\cdot)$ or $-x \cdot u''(x)/u'(x)$
- Many prominent utility functions: CARA, CRRA, HARA, ...

The Special Case of CRRA

- In Macro, we often use the CRRA utility function:

$$u(C) = \frac{C^{1-\gamma} - 1}{1-\gamma}$$

- The parameter γ is the coefficient of relative risk aversion (RRA)

$$\frac{-c \cdot u''(c)}{u'(c)} = \gamma$$

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- If $x \succeq y$, then $\alpha x \succeq \alpha y$ for any $\alpha > 0$ (Indifference curves are radially symmetric)
- Optimal portfolio weights are independent of wealth (with complete markets)
- Equilibrium prices (and returns) depend on *growth rates*, not levels

Questions?

Intertemporal Substitution

- You don't need the concept of *uncertainty* to think about intertemporal substitution
- Think about two periods and a consumer choosing optimal consumption:

$$\max_{C_1, C_2} u(C_1) + \beta u(C_2) \quad \text{s.t.} \quad C_1 + \frac{C_2}{1+r} = W$$

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- **Elasticity of Intertemporal Substitution (EIS):** $\frac{d \log(c_2/c_1)}{d \log(1+r)} > 0$
- Exercise: show that for CRRA preferences, the EIS is $1/\gamma$

Risk Aversion vs Intertemporal Substitution

- With CRRA preferences, high $\gamma \implies$ high risk aversion
- But high $\gamma \implies$ low EIS $\implies$ low willingness to substitute consumption
- Only two possibilities:
 - High γ : very risk-averse, hates shifting consumption around
 - Low γ : not very risk-averse, and very willing to shift consumption around

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 - Low γ : not very risk-averse, and very willing to shift consumption around
- If you bump up γ to make the consumer afraid of equity...
- She will really upset about shifting consumption around!
- You have to make the relative price of consuming tomorrow low $\implies$ high risk-free rate!
- The prize for waiting must be large... risk-free rate puzzle from Weil (1989)!

Questions?

A New Class of Preferences Comes to Town

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- The timing of uncertainty is irrelevant
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- With expected utility, only probabilities and outcomes matter
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- But what if you care about *when* uncertainty is resolved?
- An irreversible medical condition... do you want to find out now or later?
- At the slot machine in Vegas... do you want to wait and gamble, or to resolve it immediately?
- It's already decided whether you get tenured or not... do you want to find out now or later?

Kreps-Porteus Preferences

- Kreps and Porteus (1978): axiomatic characterization of preferences over lotteries *and* timing
- In the EU framework, “time” only matters because it induces other *states*...
- Example: flip a coin, consume 5 at $t = 0$, 10 at $t = 1$ if heads, 0 at $t = 1$ if tails
- Do you want to find out the outcome of the coin flip at $t = 0$ or at $t = 1$?

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- KP preferences allow you to prefer one over the other
- EU: preferences over probability distributions
- KP: preferences over “filtrations” – information structures
- KP recover EU when the agent does not care about timing

Quick Look Into Their Setup (Skipping Super Technical Terms)

- Time is discrete $t \in \{0, 1, 2, \dots, T\}$
- For every t , there is a payoff space Z_t
- D_T is the set of probability measures over Z_T (lotteries at time T)

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- In general: $D_t \equiv Z_t \times X_{t+1}, \forall t < T$
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- $D_{T-1} \equiv Z_{T-1} \times X_T$
- In general: $D_t \equiv Z_t \times X_{t+1}, \forall t < T$
- Your preference relation $\succeq$ ranks elements of D_t for all t
- Usual stuff: completeness, transitivity, continuity, monotonicity, ...
- If you want some stuff, check the paper!

Example

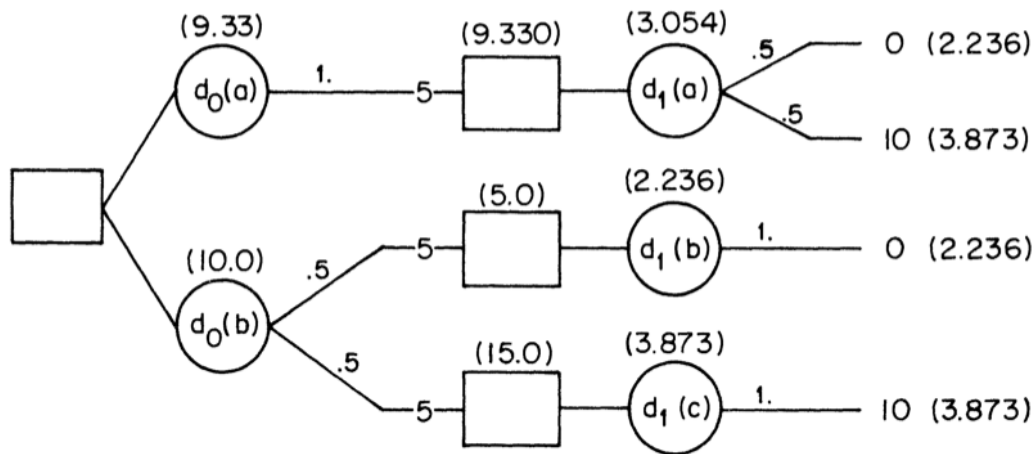


FIGURE 2.

The Representation Theorem

- In a finite decision tree, suppose U_t is your continuation value
- U_{t+1} : the *random* (from the point of view of t) continuation value tomorrow
- Let $\mu_t(U_{t+1})$ be the certainty equivalent of U_{t+1}
- Interpretation: money you get right now, for sure, to stop playing

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- Let $\mu_t(U_{t+1})$ be the certainty equivalent of U_{t+1}
- Interpretation: money you get right now, for sure, to stop playing
- Axioms from Kreps and Porteus (1978) $\implies$ there exists an “aggregator” function f such that

$$U_t = f(Z_t, \mu_t(U_{t+1}))$$

- $\mu_t(\cdot)$ encodes how risk-averse you are: the more risk-averse, the lower $\mu_t(U_{t+1})$ is
- The aggregator measures how much you trade-off payoff today vs keep playing
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- If f is linear in the second argument and $\mu_t(U_{t+1}) = \mathbb{E}_t[U_{t+1}]$ we recover expected utility!

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- They extended Kreps and Porteus (1978) to the infinite horizon case
- Super non-trivial since there is no “end date”... continuation becomes a fixed point problem

- Epstein and Zin (1989) made the whole setup operational
- They extended Kreps and Porteus (1978) to the infinite horizon case
- Super non-trivial since there is no “end date”... continuation becomes a fixed point problem
- They specialize functional forms:

$$f(x, y) \equiv ((1 - \beta)x^{1-\rho} + \beta y^{1-\rho})^{\frac{1}{1-\rho}}$$
$$\mu_t(U_{t+1}) \equiv \left(\mathbb{E}_t[U_{t+1}^{1-\gamma}] \right)^{\frac{1}{1-\gamma}}$$

Can Kreps-Porteus Preferences Solve the Equity Premium Puzzle?

- Weil (1989): no, not really. You need way more than only that

Net risk premium and net risk-free rate ^a ($\beta = 0.98$)							
EIS ($1/\rho$)	CRRA (γ)						
	0	0.5	1	5	10	20	45
∞	0.00	0.05	0.09	0.47	0.93	1.76	3.00
	2.04	2.02	1.99	1.80	1.58	1.21	0.91
2	0.01	0.06	0.11	0.51	1.00	1.88	3.13
	2.95	2.91	2.87	2.55	2.17	1.53	0.75
1	0.01	0.07	0.12	0.55	1.07	2.00	3.27
	3.86	3.81	3.75	3.31	2.77	1.85	0.59
0.2	0.10	0.18	0.26	0.89	1.65	2.93	4.37
	11.49	11.29	11.10	9.56	7.72	4.45	-0.68
0.1	0.25	0.36	0.47	1.33	2.37	4.10	5.79
	21.87	21.47	21.08	17.95	14.26	7.83	-2.24
0.05	0.59	0.75	0.89	2.18	3.73	6.37	8.82
	45.89	44.95	43.98	36.92	28.72	15.01	-5.22
1/45	1.19	1.42	1.66	3.70	6.40	11.50	17.53
	131.52	128.87	125.06	101.02	75.11	36.06	-12.00

^aIn percent The bold number is the net risk-free rate

- There are two more fundamental reasons for the puzzle:
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 - There is not enough aggregate risk in consumption growth
 - The average consumption growth is too high to be consistent with low interest rates
- Increasing risk in consumption growth: disasters, long-run risks, ...
- Or introduce very imperfect insurance markets (e.g. incomplete markets, borrowing constraints, ...)
- Heterogeneity across agents is likely important!

- Campbell, Chap. 6

Epstein, Larry, and Stanley Zin. 1989. "Substitution, Risk Aversion, and the Temporal Behavior of Consumption and Asset Returns: A Theoretical Framework." *Econometrica* 57 (4): 937–69.

<https://EconPapers.repec.org/RePEc:ecm:emetrp:v:57:y:1989:i:4:p:937-69>.

Kreps, David M., and Evan L. Porteus. 1978. "Temporal Resolution of Uncertainty and Dynamic Choice Theory." *Econometrica* 46 (1): 185. <https://doi.org/10.2307/1913656>.

Weil, Philippe. 1989. "The Equity Premium Puzzle and the Risk-Free Rate Puzzle." *Journal of Monetary Economics* 24 (3): 401–21. [https://doi.org/10.1016/0304-3932\(89\)90028-7](https://doi.org/10.1016/0304-3932(89)90028-7).